



Ahmedabad
University

CSE623 : Machine Learning

**Machine Learning-based Estimation of Muscle
Cross-Sectional Area (ACSA) from Segmented
Ultrasound Images of Lower Limb Muscles for
Sports Performance Analysis**

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Abstract

This work proposes a machine learning-based pipeline to classify the Rectus Femoris (RF) muscle in ultrasound images and predict Anatomical Cross-Sectional Area (ACSA). The method involves preprocessing with thresholding, cropping, a median filter, and CLAHE, as well as pixel-wise intensity, texture, gradient, and spatial feature extraction. Hard-negative sampling is used in a Random Forest classifier to achieve pixel-level classification of the muscle and background regions. The model has a validity score of 0.726 and a test score of 0.6968, and its pixels are highly accurate (approximately 0.94), meaning that it works on larger regions very well but has problems with edge accuracy. The pixel-to-area conversion factor is calibrated to estimate areas automatically, a characteristic of ACSA. The further research will focus on enhancing feature representations, boundary refinement, and post-processing methods, as well as on model robustness, to increase the accuracy and reliability of segmentation.

Contents

1	What We Have Done	4
2	What We Aspire to Do	6

1 What We Have Done

- We developed a well working pipeline for segmentation of Rectus Femoris(RF) muscle regions from the ultrasound image dataset provided for estimation of ACSA. The dataset was divided into training, test and validation containing 236 images, 52 images and 50 images respectively.
- The images were preprocessed in correspondence to the ground-truth masks provided and were converted into matrix representations for pixel-level operations. The segmentation masks were further represented in binary form to distinguish muscle regions from the background.
- A series of preprocessing methods was applied to normalise the images and reduce noise, including thresholding, cropping the region of interest, median filtering, and CLAHE.
- After pre-processing, pixel-wise features such as intensity, texture, gradient and spatial characteristics were extracted.
- We used a Random Forest classifier on the extracted features to predict the type of pixel, muscle or background (1 or 0). Morphological processes like closing, hole filling and removal of small connected components were used to refine the predicted segmentation masks.
- Post-processing included techniques such as morphological cloning, hole filtering and removal of small connected components, which were applied to refine the segmentation masks and improve structural consistency.
- Model performance was evaluated using segmentation metrics, including Dice score and IoU. Our model achieved a Dice score of 0.726 and an IoU of 0.590 on the validation set, and a Dice score of 0.6968 and an IoU of 0.5563 on the test set.
- Lastly, we calculated the Anatomical Cross-Sectional Area (ACSA) by transforming the number of pixels that have been segmented into physical area using a known pixel-to-area constant, which allows estimation of muscle area automatically using ultrasound imaging.

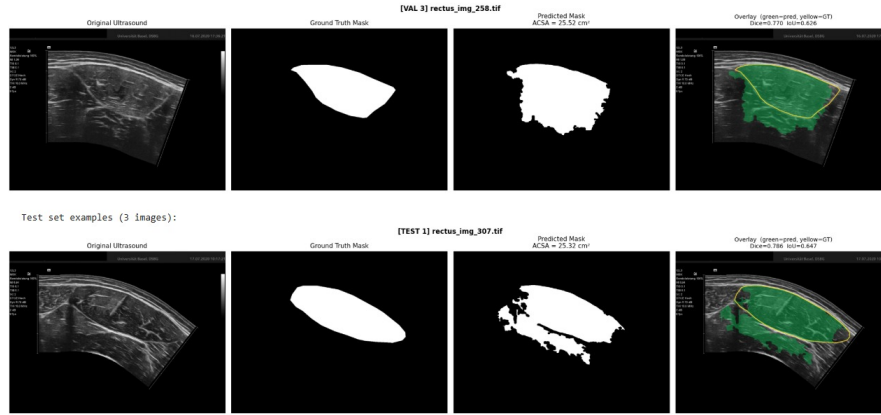


Figure 1: Qualitative Segmentation Results on Validation and Test Samples with ACSA Estimation and Overlay Comparison

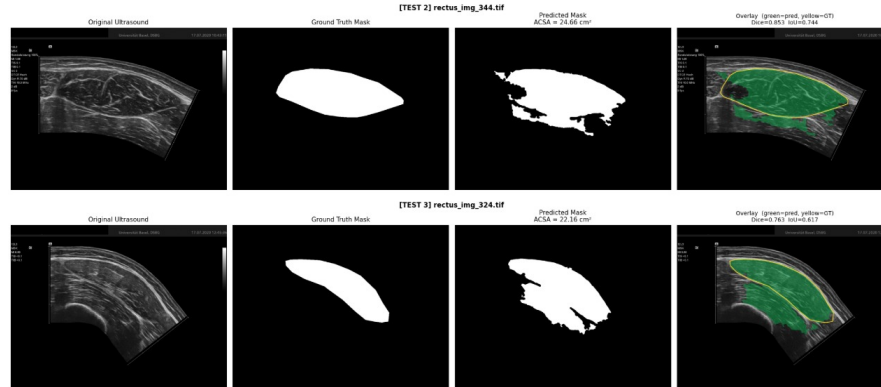


Figure 2: Qualitative Segmentation Results Showing Boundary Variations and Prediction Differences (Test Sample)

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PIPELINE COMPLETE - v5
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Split : 236 train | 50 val | 52 test

VAL → Dice 0.7262 | IoU 0.5901 | Acc 0.9456
TEST → Dice 0.6968 | IoU 0.5563 | Acc 0.9441

ACSA median calibration : 0.000189 cm²/pixel

To predict on a new image:
result = predict_new_image('scan.tif', clf, median_cal)
print(result['acsa_cm2'], 'cm²')
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Figure 3: Qualitative Segmentation Results: Original Ultrasound Image, Ground Truth Mask, Predicted Mask, and Overlay Comparison (Test Sample)

2 What We Aspire to Do

- In order to have an even better segmentation performance, we are interested in improving the existing machine learning pipeline with the addition of more discriminative and robust features representations. This encompasses texture, gradient, and, to a greater extent, multi-scale features that are better able to reflect the structural changes in ultrasound images.
- Special attention will be devoted to the enhancement of the precision of boundaries, since the existing findings are unsatisfactory, marked by coarse edges and segmentation leakage in difficult areas. This will be addressed with boundary-aware feature engineering, enhanced sampling plans around edges and enhanced post-processing methods.
- We will include more recent data augmentation methods, such as elastic deformations, contrast enhancement, and noise simulation, to further improve the model's resilience and extrapolation, especially in low-contrast regions where speckle noise is present.
- The multi-scale method of feature extraction will be further developed so as to capture more of the global structure and local details that are vital in the precise segmentation of the thin and irregular regions.
- Random Forest model will also be optimised using hyperparameters. tuning, which involves the number of trees, the maximum depth, and feature selection strategies and better sampling methods to take better care in terms of class imbalance and boundary regions.
- The adaptive morphological operation, connected component analysis, and contour refining will be included in the post-processing methods to generate smoother and more naturalistic segmentation masks.
- To enhance the accuracy of the ACSA estimation, we will optimise the calibration operation and eliminate segmentation noise by enhancing the predictive stability and post-processing. We will also examine how segmentation accuracy is associated with ACSA estimation to ensure greater consistency and clinical relevance. meaningful results.
- Lastly, additional evaluation metrics, such as the Hausdorff Distance and precision/recall analysis, will be added to provide further insight into the quality of the boundaries and model performance in severe areas.

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